Linear Regression Part 2

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$x_1 \; x_2 \; \cdots \; x_n$ n observatio

$$\begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \quad Y \quad \hat{\beta}$$

$$\hat{\beta} = (X'X)^{-1} X'Y$$

How good these estimates are

Metrics → Sum of Squares

Metrics → Degrees of freedom

Topics

- Degrees of Freedom
- Residual Variance
- Resolution of Sum of Squares
- Finding R^2 and Adjusted R^2 PRESS
- Variance-Covariance Matrix Very Imp.

Actual response and its prediction

Express the deviation between an actual response (Y_i) and the predicted response ($\hat{Y}_i$) in terms of difference of two deviations

- Deviation of actual response from mean response $\bar{Y}$:

$$Y_i - \bar{Y}$$

- Deviation of predicted response from mean response:

$$\hat{Y}_i - \bar{Y}$$

How to Find Mean Response?

$$\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n}$$

Interesting Aside: It may be shown that the mean of the predicted responses is the same as the mean of the actual (experimental) responses

$$\bar{Y} = \frac{\sum_{i=1}^{n} \hat{Y}_i}{n} = \bar{\hat{Y}}$$

Relation between actual response and its prediction

- Deviation of actual response from mean response
- Deviation of predicted response from mean response

$$(Y_i - \overline{Y}) = (Y_i - \hat{Y}_i) + (\hat{Y}_i - \overline{Y})$$

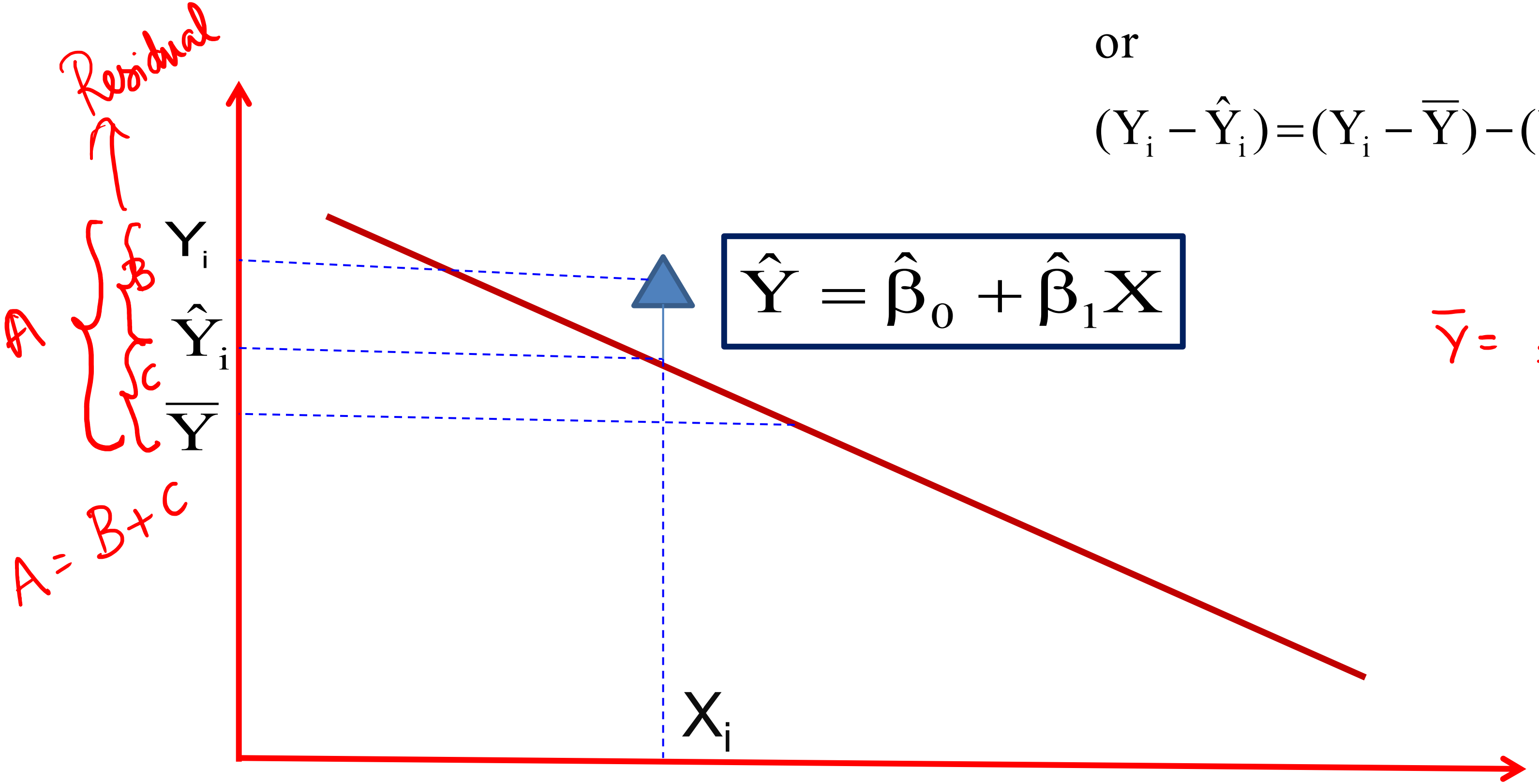
or

$$(Y_i - \hat{Y}_i) = (Y_i - \overline{Y}) - (\hat{Y}_i - \overline{Y})$$

$$(Y_i - \overline{Y}) = (Y_i - \hat{Y}_i) + (\hat{Y}_i - \overline{Y})$$

or

$$(Y_i - \hat{Y}_i) = (Y_i - \overline{Y}) - (\hat{Y}_i - \overline{Y})$$



$$\overline{Y} = \frac{\sum_{i=1}^{n} Y_i}{n}$$

Sum of Squares

It may be shown that

$$\sum_{i=1}^{n}(Y_i-\overline{Y})^2 = \sum_{i=1}^{n}(Y_i-\hat{Y}_i)^2 + \sum_{i=1}^{n}(\hat{Y}_i-\overline{Y})^2$$

$$\sum_{i=1}^{n}(Y_i-\overline{Y})^2 = \sum_{i=1}^{n}(Y_i-\hat{Y}_i)^2 + \sum_{i=1}^{n}(\hat{Y}_i-\overline{Y})^2$$

$$\underbrace{SS_{Total}}_{\text{Unchanged ?}} \quad \underbrace{SS_{Residual}}_{\text{minimize ?}} \; \text{reduce} \quad \underbrace{SS_{Regression}}_{\text{maximize ?}} \; \text{increase}$$

Add more parameters to model

This amounts to

Sum of Squares of the Deviations of the Responses from the Mean =

Sum of Squares of the Residuals + Sum of Squares due to Regression

Degrees of Freedom (DOF)

Sum of Squares ✓

Broadly, let there be a group of numerical entities X_1, X_2, …,X_n.

Suppose we use these to find the mean value.

$\bar{x} = \sum_{1}^{n} x_i / n$

Sample: x_1, x_2, …, x_n

Mean value of population μ Not known

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Now, one relation, viz. mean(or average) has been found from these n numbers.

Sum of Deviations from the Mean

3 deviations from mean

$n-1$

$d_1 \quad d_2 \quad d_3$

$d_1 + d_2 + d_3 = 0$

2 dof

$d_3 = -d_1 - d_2$

$d_1 + d_2 + d_3 = 0$

Let us define deviation from mean (d_i) as

$$d_i = X_i - \bar{X} = \frac{\sum x_i}{n}$$

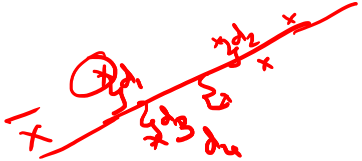
What is $\sum_{i=1}^{n} d_i$?

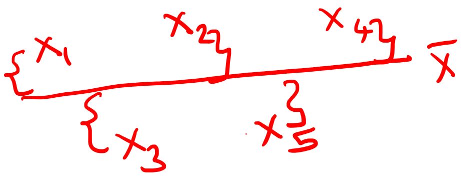
$= 0$

x_i	d_i
0	-1
1	0
2	1
Mean 1	0

	d_i
2	-2
4	0
6	2
Mean 4	0

Sum of 2 deviations is 2. What is the third deviation? -2





Sum of Deviations from the Mean

Let us define deviation from mean (d_i) as

$$d_i = X_i - \bar{X}$$

$$\sum_{i=1}^{n} d_i = \sum_{i=1}^{n} X_i - \sum_{i=1}^{n} \bar{X}$$

$$\bar{x} = \frac{\sum_{i=1}^{n} X_i}{n}$$

$$\sum_{i=1}^{n} d_i = n\bar{X} - n\bar{X} = 0$$

Hence, sum of deviations equals zero.

Sum of Deviations from the Mean

Hence, sum of deviations equals zero

$$\sum_{i=1}^{n} d_i = 0$$

$$\underbrace{d_1 + d_2 + \ldots +}_{i-1} \underline{\underline{d_i}} \underbrace{+ \cdots + d_n}_{n-i} = 0$$

Here there are only n-1 independent deviations as the i^{th} deviation d_i may be found as follows

$$d_i = -(d_1 + d_2 + d_{i-1} + d_{i+1} + \cdots + d_n)$$

Sample Standard Deviation S^2

Hence, the standard deviation of the sample is defined as

n-1 indep. deviations

Sample Variance has n-1 dof

$$S^2 = \frac{\sum_{i=1}^{n} d_i^2}{n-1} = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}$$

popln mean

$$\frac{\sum_{i=1}^{n} (X_i - \mu)^2}{n}$$

Sample entity

S^2 is based upon using the sample mean in its definition which in turn is based on only n-1 independent deviations from the sample mean.

Sample Standard Deviation S^2

$$S^2 = \frac{\sum_{i=1}^{n} d_i^2}{n-1} = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}$$

- The sum of squares of the deviations from the mean $\sum_{i=1}^{n} d_i^2$ involves only n-1 independent squared deviations.
- Hence, there are only n-1 degrees of freedom among the n d_i^2 due to the constraint $\sum_{i=1}^{n} d_i = 0$

Degrees of Freedom

$$SS_{Total} = + SS_{Residual} + SS_{Regression}$$

$$\sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 + \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$$

DOF: n-1 n-p + k (=p-1)

The SS_{Total} has n-1 degrees of freedom (HOW?)

The $SS_{Regression}$ has k degrees of freedom where k is related to total number of regression coefficients as

$\frac{\partial L}{\partial \hat{\beta}_0} = 0$ $p = 2$ eqns $n-2$

$\frac{\partial L}{\partial \hat{\beta}_1} = 0$ 2 Constraints

$\frac{\partial L}{\partial \hat{\beta}_3} = 0$ $p = 3$ eqns $n-3$

3 Constraints

Objective Loss Fn.

$L \rightarrow \hat{\beta}_0$, $L \rightarrow \hat{\beta}_1$

$\hat{\beta}_0 + \hat{\beta}_1 x_1$ $p = 2 = 1 + 1$

$\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_1^2$ $p = 3 = 1 + 2$

$3 = 1 + k$

$(\cancel{n} - 1) = (\cancel{n} - p) + k$?

$p = k + 1$

$\underbrace{\hat{\beta}_0, \hat{\beta}_1, \dots \hat{\beta}_k}$

$p = 1 + k$

Omitting β_o

- Hence, it may be seen that the parameter β_o has been omitted from the regression sum of square and
- Only the partial regression coefficients β_1, $\beta_{2,}$ … β_k have been included

Why n-p in $SS_{Residual}$?

$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$

$$\frac{\sum_{i=1}^{n} {e_i}^2}{n-p} = \frac{SS_{Residual}}{n-p} = \text{Residual Variance}$$

$$SS_{Residual} = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$

$$SS_{Residual} = \sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2$$

- Two parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ are estimated here from the data.
- Two constraints operate on the sum of squares.
- The constraints operating on $\sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2$ are

$$\frac{\partial SS_{Residual}}{\partial \hat{\beta}_0} = \frac{\partial}{\partial \hat{\beta}_0}\left[\sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2\right] = 0 \qquad \frac{\partial SS_{Residual}}{\partial \hat{\beta}_1} = \frac{\partial}{\partial \hat{\beta}_1}\left[\sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2\right]$$

Why n-p?

- Two parameters are estimated here from the data. Two constraints operate on the sum of squares.
- The constraints operating on $\sum_{i=1}^{n}\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2$ are

$$\frac{\partial SS_E}{\partial \hat{\beta}_0} = \frac{\partial}{\partial \hat{\beta}_0}\left[\sum_{i=1}^{n}\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2\right] = 0$$

$$\frac{\partial SS_E}{\partial \hat{\beta}_1} = \frac{\partial}{\partial \hat{\beta}_1}\left[\sum_{i=1}^{n}\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right)^2\right] = 0$$

Analysis of Variance (ANOVA)

Source of Variation	Sum of Squares	Degrees of Freedom
Regression	$SS_{Regression}$	$k = p-1$
Residual	$SS_{Residual}$	$n-p$
Total	SS_{Total}	$n-1$

$\sum (Y_i - \bar{Y})^2$

$= \sum Y_i^2 - \underbrace{n\bar{Y}^2}_{\text{Contribution by } \beta_0}$

$\sum_{i=1}^{n} Y_i^2 - 2\sum_{i=1}^{n} Y_i \bar{Y} + \sum_{i=1}^{n} \bar{Y}^2$

$\sum Y_i^2 - 2n\bar{Y}^2 + n\bar{Y}^2$

$= \sum Y_i^2 - n\bar{Y}^2$

Source of Variation	Sum of Squares	Degrees of Freedom
Regression	$SS_{Regression}$	k
Residual	$SS_{Residual}$	n-p
Total	SS_{Total}	n-1

Here n is the total number of observations and p is the total number of parameters (including the constant β_o).

$$\text{Since } k = (n-1) - (n-p)$$

$$\mathbf{k = p-1}$$

Definition of Total Sum of Squares

The total sum of squares is defined as

$$\mathrm{SS}_{\mathrm{total}} = \sum_{\mathrm{i}=1}^{\mathrm{n}} (\mathrm{Y}_{\mathrm{i}} - \overline{\mathrm{Y}})^2$$

Derivation of Total Sum of Squares

$$SS_{total} = \sum_{i=1}^{n}(Y_i - \overline{Y})^2 \qquad = \sum_{i=1}^{n} Y_i^{\,2} - \frac{\left(\sum_{i=1}^{n} Y_i\right)^2}{n}$$

$$= \mathbf{Y'Y} - \frac{\left(\sum_{i=1}^{n} Y_i\right)^2}{n}$$

$$SS_{Total} = \mathbf{Y'Y} - n\overline{Y}^2$$

Derivation for Actual Regression Sum of Squares

$$SS_{Residual} = \mathbf{e'e}$$

$$= (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})$$

$$= (\mathbf{Y'Y} - 2\hat{\boldsymbol{\beta}}'\mathbf{X'Y} + \hat{\boldsymbol{\beta}}\mathbf{X'X}\hat{\boldsymbol{\beta}})$$

$$SS_{Residual} = \underbrace{(Y'Y - n\bar{Y}^2)}_{SS_{Total}} - \underbrace{(\hat{\beta}'X'Y - n\bar{Y}^2)}_{SS_{Regression}}$$

$$= (\mathbf{Y'Y} - 2\hat{\boldsymbol{\beta}}'\mathbf{X'Y} + \hat{\boldsymbol{\beta}}'\mathbf{X'Y})$$

$$SS_{Residual} = (\mathbf{Y'Y} - \hat{\boldsymbol{\beta}}'\mathbf{X'Y}) \quad \text{As} \quad \mathbf{X'X}\hat{\boldsymbol{\beta}} = \mathbf{X'Y}$$

$$SS_{Residual} = \text{Actual } SS_{Total} + \text{Actual } SS_{Regression}$$

$$-n\bar{Y}^2 \qquad\qquad -n\bar{Y}^2$$

Manipulation of Residual Sum of Squares

$$SS_{Residual} = SS_{total} - SS_{regression}$$

We previously defined $SS_{Residual}$ as follows

$$SS_{Residual} = (Y'Y - \hat{\beta}'X'Y)$$

$$SS_{Residual} = Y'Y - SS_{Regression}$$

$$\sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

$$= \sum Y_i^2 - n\bar{Y}^2$$

Significance??

Subtracting $n\bar{Y}^2$ from both terms on RHS we get

$$SS_{Residual} = (\underline{Y'Y} - n\bar{Y}^2) - (\hat{\beta}'X'Y - n\bar{Y}^2)$$

Why not $n\bar{Y}^3$?

Resolution of Error Sum of Squares

$$SS_{Residual} = (Y'Y\text{-}\hat{\beta}'X'Y)$$

$$SS_{Residual} = \left(Y'Y\text{-}\frac{\left(\sum_{i=1}^{n} Y_i\right)^2}{n}\right) - \left(\hat{\beta}'X'Y\text{-}\frac{\left(\sum_{i=1}^{n} Y_i\right)^2}{n}\right)$$

$$SS_{Residual} = SS_{total} - SS_{regression}$$

By definition of the total sum of squares, the second term becomes the regression sum of squares

Manipulation of Residual Sum of Squares

$-\underbrace{n\bar{Y}^2}$ ✓

$$SS_{Residual} = (\mathbf{Y'Y} - n\bar{Y}^2) - (\hat{\boldsymbol{\beta}}'\mathbf{X'Y} - n\bar{Y}^2)$$

$n\bar{Y}^2 = SS_{\beta_0}$

SS_{β_0}

$$SS_{Residual} =$$

(Actual SS_{Total} $\underbrace{\text{corrected for } \beta_o}$) - ($SS_{Regression}$ corrected for β_o)

Hence sum of squares due to β_o is given by $n\bar{Y}^2$

Total Sum of Squares

Total Sum of Squares: $\mathbf{Y'Y - \frac{\left(\sum_{i=1}^{n} Y_i\right)^2}{n}}$

The *actual* total sum of squares is **Y′Y.**

The term $n\bar{Y}^2 = \frac{\left(\sum_{i=1}^{n} Y_i\right)^2}{n}$ is deducted from the *actual* total sum of squares.

Regression Sum of Squares due to Partial Regression Coefficients

- The regression sum of squares does NOT include the intercept β_0 contribution and
- Has contributions only from $\beta_1, \beta_2, \beta_3, \ldots \beta_k$.
- Hence the degrees of freedom for this regression sum of squares is k (= p-1).

Actual SS contribution by Y

$\sum Y_i^2 = Y_1^2 + Y_2^2 + \cdots + Y_n^2$

Why $SS_{\beta_0} = n\bar{Y}^2$?

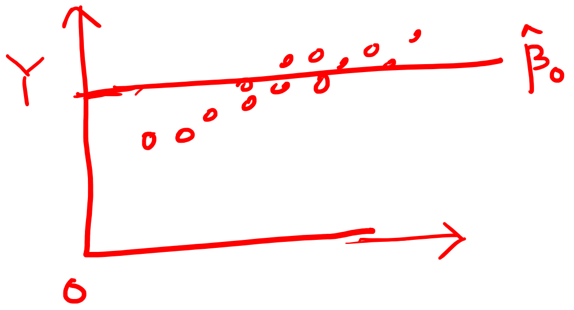
because $\beta_0 = \bar{Y}$ if other regression coeffs are zero

Regression Sum of Squares due to β_0

Another way of looking at this is as follows

If we fit a model containing only β_0 , the fitted model is

$$\hat{Y} = \bar{Y}$$



or

$$\hat{\beta}_0 = \bar{Y}$$

There is no contribution from the predictors

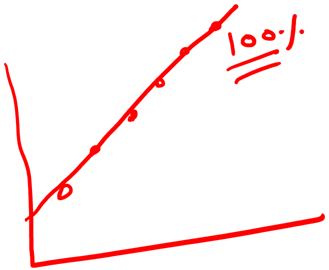
Regression Sum of Squares due to β_0

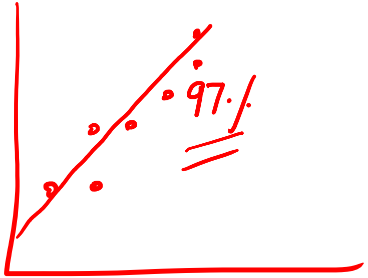
$$\hat{\beta}_o = \bar{Y}$$

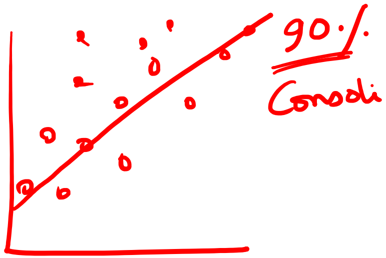
There is no contribution from the predictors

Hence sum of squares due to $\hat{\beta}_o$ is

$$\sum_{i=1}^{n} \bar{Y}^2 = n\bar{Y}^2$$







Consolidated data into

X matrix

β vector

Y vector

$Y = X\beta + \varepsilon$

$\hat{\beta} = (X'X)^{-1}(X'Y)$

$\hat{Y} = X\hat{\beta}$

Metrics of Regression :

How Good is my Fit?

SSQ dof

$\sum(Y_i-\bar{Y})^2$ $n-1$

$\sum(Y_i-\hat{Y}_i)^2$ $n-p$ +

$\sum(\hat{Y}_i-\bar{Y})^2$ $k=p-1$

$\frac{}{n-1}$

Sum of Squares

$$\sum_{i=1}^{n} Y_i^2 - n\bar{Y}^2 = \sum_{i=1}^{n}(Y_i-\hat{Y}_i)^2 + \sum_{i=1}^{n}(\hat{Y}_i-\bar{Y})^2$$

Regression SSQ

$$Y'Y - n\bar{Y}^2 = (Y-\hat{Y})'(Y-\hat{Y})' + \hat{\beta}'X'Y - n\bar{Y}^2$$

Residual SSQ

Coefficient of Determination (R^2):

Total SSQ = Residual SSQ + Regression SSQ

$Y'Y - n\bar{Y}^2$ $\sum_1 (Y_i - \hat{Y}_i)^2$ $\sum_1 (\hat{Y}_i - \bar{Y})^2$

% of variation in data described by model $= \% \; R^2 = \dfrac{SS_{\text{Regression}}}{SS_{\text{Total}}} \times 100$

100   25 + 75

$R^2 = 0.75$

This represents the proportion of the total variability accounted or explained by the linear regression model

Total variation in data = SS_{Total} fixed

Variation Actually explained by model = $SS_{Regression}$

Variation unaccount = $SS_{Residual}$

Coefficient of Determination (R^2):

$\neq$ corr'ln coeff.

The R^2 value may be increased by increasing the number of terms in the model and thereby increasing the number of coefficients that may be adjusted so that the model can be made to fit the data excellently.

Coefficient of Determination (R^2):

- A complex model becomes cumbersome to handle, more empirical in nature and difficult to ascribe physical reasons for the dependence of the response on the factors of the experiment.
- A noisy data fitted with too many parameters will fail to work well when implemented.

Adjusted R^2:

Here Mean Squares of the error and model/total sum of squares are used.

$$R^2_{adj} = 1 - \frac{SS_{Error}/(n-p)}{SS_{Total}/(n-1)}$$

Adjusted R^2:

$$R^2_{adj} = 1 - \frac{SS_{Error} / (n-p)}{SS_{Total} / (n-1)}$$

To penalize the addition of unnecessary terms in the model equation ("overfitting") the adjusted R^2 value also gets reported.

Adjusted R^2:

$$R^2_{adj} = 1 - \frac{SS_{Error} / (n-p)}{SS_{Total} / (n-1)}$$

The SS_{Error} contribution is increased by dividing with the error degrees of freedom. The latter reduces when more number of parameters are taken up by the model.

Adjusted R^2:

$$R^2_{adj} = 1 - \frac{SS_{Error}/(n-p)}{SS_{Total}/(n-1)}$$

Unless the SS_{Error} is considerably reduced by adding the extra term to the model equation, the adjusted R^2 will increase upon increasing the number of parameters.

PRESS: Prediction Error Sum of Squares

- This term is also similar to the sum of squares of the residuals.
- We sum the square of the deviations between the actual responses and the corresponding model predicted values.

PRESS: Prediction Error Sum of Squares

The main difference to watch out for is that the prediction for the i^{th} data is based on a model equation that *excluded* that particular data point but used the remaining data points to develop the model equation.

PRESS: Prediction Error Sum of Squares

- The same treatment is meted to other data points as well when they are compared to their individual model predictions.

PRESS:

First **H** matrix is defined as follows

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

Let us define the usual residual

$$\mathrm{e_i} = (\mathrm{y_i} - \hat{\mathrm{y}}_\mathrm{i})$$

The PRESS residual $\mathrm{e_{(i)}}$ may be related to $\mathrm{e_i}$ as

$$\mathrm{e_{(i)}} = \mathrm{e_i}/(1 - \mathrm{h_{ii}})$$

Here, $\mathrm{h_{ii}}$ refers to the diagonal elements of **H**.

PRESS:

The PRESS residual $\mathrm{e}_{(\mathrm{i})}$ may be related to e_i as

$$\mathrm{e}_{(\mathrm{i})} = \frac{\mathrm{e}_\mathrm{i}}{1 - \mathrm{h}_{\mathrm{ii}}}$$

Here, h_{ii} refers to the diagonal elements of **H**

$$\mathrm{PRESS} = \sum_{\mathrm{i}=1}^{\mathrm{n}} \left(\frac{\mathrm{e}_\mathrm{i}}{1 - \mathrm{h}_{\mathrm{ii}}} \right)^2$$

$$\mathrm{R}^2_{\mathrm{PRESS}} = 1 - \frac{\mathrm{PRESS}}{\mathrm{SS}_{\mathrm{Total}}}$$

Remember to adjust for intercept when finding total sum of squares

Variance-Covariance Matrix

The variance-covariances of the least square estimators of the parameters $\hat{\beta}$ including $\hat{\beta}_0$ are the elements of the **$(X'X)^{-1}$** matrix multiplied by the variance σ^2.

$(X'X)^{-1}$ σ^2 is termed as the

Variance-Covariance matrix.

Variance-Covariance Matrix

The diagonal elements of the Covariance matrix are the variances of the least square estimators and the off-diagonal elements are the covariances.

The Covariance matrix dimensions are p x p

Form of the Variance-Covariance Matrix

$$\mathbf{C} = (\mathbf{X'X})^{-1}\sigma^2 = \begin{bmatrix} C_{00} & C_{01} & C_{02} \\ C_{10} & C_{11} & C_{12} \\ C_{20} & C_{21} & C_{22} \end{bmatrix}\sigma^2$$

Now $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is a matrix of constants, and the variance of $\mathbf{y}$ is $\sigma^2\mathbf{I}$, so

$$\begin{aligned} \mathrm{Var}(\hat{\boldsymbol{\beta}}) &= \mathrm{Var}\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}\right] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathrm{Var}(\mathbf{y})\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right]' \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

Therefore, if we let $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}$, the variance of $\hat{\beta}_j$ is $\sigma^2 C_{jj}$ and the covariance between $\hat{\beta}_i$ and $\hat{\beta}_j$ is $\sigma^2 C_{ij}$.

Matrix Form of the Regression Equations

This is a symmetric matrix

($C_{10}=C_{01}$, $C_{20}=C_{02}$, and $C_{21}=C_{12}$) as $\mathbf{(X'X)^{-1}}$ is symmetric.

$$V(\hat{\beta}) = \sigma^1 C_{jj}, j = 0,1,2$$

$$Cov(\hat{\beta}_i, \hat{\beta}_j) = \sigma^1 C_{ij}, i, j = 0,1,2, i \neq j$$

Significance of the Covariance Matrix

The estimates of the variances of these estimated regression coefficients are obtained by replacing σ^2 with an estimate $\hat{\sigma}^2$ as σ^2 is unknown.

Estimated Standard Error

When σ^2 is replaced by its estimate $\hat{\sigma}^2$, the square root of the estimated variance of the j^{th} regression coefficient is called the **estimated standard error** of the least squares estimator $\hat{\beta}_j$

Significance of the Covariance Matrix

The **estimated standard error** of the least squares estimator is given by

$$se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$$

Significance of the Covariance Matrix

- These standard errors are a measure of the precision of the estimation for the regression coefficients
- Small standard errors imply good precision.

Estimator of Standard Error

- The estimation of residual error is important
- The residual should ideally reflect the difference due to random factors and not systematic discrepancies created by using an inadequate model.
- This is an unbiased estimator of σ^2

Estimator of Residual Variance

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} e_i{}^2}{n-p} = \frac{SS_E}{n-p}$$

$$SS_E = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$

Standard Error

$$se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$$

Compare the standard errors $se(\hat{\beta}_j)$ with the estimated coefficients $\hat{\beta}_j$ and if the standard errors are much smaller than these coefficients, **the precision** of estimation is reasonably good.

Tutorial Problem

For the conveyor belt problem find

a) R^2

b) Adjusted R^2

c) PRESS

d) Variance – Covariance Matrix

e) Standard Error of the Estimated Parameters

Relevant Results

For the conveyor belt problem find

a) R^2

b) Adjusted R^2

c) PRESS

d) Variance – Covariance Matrix

e) Standard Error of the Estimated Parameters

a) Coefficient of Determination R^2

Previously the total sum of squares and regression sum of squares were actual values viz.

$$\text{Actual } SS_{Total} = \sum_{i=1}^{n} Y_i^2 \;\&\; \text{Actual } SS_{Regression} = \beta' X' Y$$

But we need to use

$$SS_{Total} = \sum_{i=1}^{n} Y_i^2 - n\overline{Y}^2 \quad \& \quad SS_{Regression} = \sum_{i=1}^{n} \beta' X' Y - n\overline{Y}^2$$

If you may recall, we are removing the effect of the intercept β_o and focussing on the partial regression coefficients.

a) Coefficient of Determination (R^2):

$$R^2 = \frac{SS_{Regression}}{SS_{Total}}$$

From Matlab we get the following results for the II order model

$$R^2 = \frac{SS_{Regression}}{SS_{Total}}$$

a) Coefficient of Determination R^2

Previously the total sum of squares and regression sum of squares were actual values viz.

$$\text{Actual } SS_{Total} = \sum_{i=1}^{n} Y_i^2 \;\; \& \;\; \text{Actual } SS_{Regression} = \beta' X' Y$$

But we need to use

$$SS_{Total} = \sum_{i=1}^{n} Y_i^2 - n\overline{Y}^2 \quad \& \quad SS_{Regression} = \sum_{i=1}^{n} \beta' X' Y - n\overline{Y}^2$$

If you may recall, we are removing the effect of the intercept β_o and focussing on the partial regression coefficients.

```
% For II order model find Total and Regression Sum of Squares

SS_Total = Y'*Y-numel(Y)*mean(Y)^2;             % 228.9091

SS_Regression = beta_hat'*X'*Y-numel(Y)*mean(Y)^2;            % 221.6775

% numel(Y) = 11 value of n

n=numel(Y);           % n=11

R2_IIorderModel = SS_Regression /SS_Total; % R2 = 0.9684
```

a) Coefficient of Determination (R^2):

$$R^2 = \frac{SS_{Regression}}{SS_{Total}}$$

From Matlab we get the following results for the II order model

$$R^2 = \frac{221.6775}{228.9091} = 0.9684$$

b) Adjusted R^2

```
% Next to find Adjusted R^2

p = numel (beta_hat) % 3

R2Adj_IIoderModel = 1 - (RsSSQ1/(n-p))/(SS_Total/(n-1));

% R2_Adj = 0.9605
```

$$\mathrm{Adj.\,R^2} = 1 - \frac{\frac{\mathrm{SS_{Residual}}}{n-p}}{\frac{\mathrm{SS_{Total}}}{n-1}}$$

$$\mathrm{Adj.\,R^2} = 1 - \frac{\frac{7.2316}{11-3}}{\frac{228.9091}{11-1}} = 0.9605$$

c) PRESS

First **H** matrix is defined as follows

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

Let us define the usual residual

$$e_i = (y_i - \hat{y}_i)$$

The PRESS residual $e_{(i)}$ may be related to e_i as

$$e_{(i)} = e_i/(1 - h_{ii})$$

Here, h_{ii} refers to the diagonal elements of **H**.

c) PRESS:

The PRESS residual $e_{(i)}$ may be related to e_i as

$$e_{(i)} = \frac{e_i}{1 - h_{ii}}$$

Here, h_{ii} refers to the diagonal elements of **H**

$$\text{PRESS} = \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}} \right)^2$$

$$R^2_{\text{PRESS}} = 1 - \frac{\text{PRESS}}{SS_{\text{Total}}}$$

Remember to adjust for intercept when finding total sum of squares

c) PRESS:

```
% To find PRESS
% First find H Matrices
HX=X*(inv(X'*X))*X';

PRESS_X=0.0;
Res_SSQ2_X=0;

for i=1:numel(Y)
Res_SSQ2_X=Res_SSQ2_X+(Y(i)-Y_Pred(i))^2;

% PRESS Calculations
PRESS_X = PRESS_X + ((Y(i)-Y_Pred(i))/(1-HX(i,i)))^2;
% PRESS = 11.9361  Predicted Sum of Squrares
end
```

```
% Find R2 PRESS

R2_PRESS=1-PRESS_X/(Y'*Y-n*(mean(Y))^2);

% R2_PRESSX = 0.9479
```

d) Variance Covariance Matrix

```
VarCov_Matrix = (inv(X'*X))*(RsSSQ1/(n-p));

% VarCov_Matrix =

%

% 40.1613 -5.6542  0.1898

% -5.6542   0.8061 -0.0273

% 0.1898   -0.0273  0.0009

% Note the symmetric nature of the Variance-Covariance Matrix
```

e) Standard Errors of the Estimated Parameters

```
VarCov_Matrix = (inv(X'*X))*(RsSSQ1/(n-p));

% RsSSQ1/(n-p) = 0.9039

% Get the diagonal elements

Var_BetaHat = diag(diag(VarCov_Matrix));

% Var_BetaHat =

%

% 40.1613        0          0

% 0           0.8061        0

% 0              0        0.0009
```

```
StdError_Beta=sqrt(Var_BetaHat);

% StdError_Beta =

%

% 6.3373        0          0

% 0          0.8979        0

% 0             0        0.0306
```